More than 67.3% of the zeros of the Riemann zeta function are simple and lie on the critical line

Abstract

Let $N(T,2T)$ count the zeros of the Riemann zeta function with ordinates in $(T,2T]$, with multiplicity, and let $N_0^s(T,2T)$ count the simple zeros on the critical line in that interval. We refine the rank–trace argument of Claude [1] and prove

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \geq 0.673008527927\ldots.$$

The proof uses the analytic estimates of Theorem D in [1] and a rigorously verified seven-point inequality for the simple-zero vectors.

1 Statement and setup

We use the notation and normalisations of [1, §§2,4,7.1]. Put

$$\ell := \log\frac{T}{2\pi}, \qquad I := (T,2T], \qquad D_0 := T^{1/2}, \qquad I' := (T-D_0, 2T+D_0].$$

For the optimised test family of Theorem D, $L=\ell$ and

$$\phi(u) = \cos\left(\frac{\sqrt{2}\,u}{\ell}\right)^{1/2} \varrho(L/2-|u|),$$

where the fixed-width ramp ϱ is defined in [1, §2.2]. Let

$$a := \frac{1}{L}\int_{\mathbb{R}} \phi(u)^2\,du, \qquad \widehat{G} := \frac{G}{aL^2}, \qquad \widehat{A} := \frac{A}{aL^2}, \qquad d := \left\lfloor \frac{LT}{2\pi} \right\rfloor.$$

As in [1, §4.1], divide the zeros in I' into the simple critical-line zeros $\mathcal{S}_1$, the distinct multiple critical-line zeros $\mathcal{S}_2$, and the unordered off-line pairs $\mathcal{P}$. Write

$$s_1 := \#\mathcal{S}_1, \qquad s_2 := \#\mathcal{S}_2, \qquad p := \#\mathcal{P}.$$

Then

$$N(I') \geq s_1 + 2s_2 + 2p. \tag{1}$$

For $\rho \in \mathcal{S}_1$, define

$$u_\rho := \big(\widehat{\phi}(\gamma_\rho - \tau_k)\big)_{0\leq k<d} \in \mathbb{R}^d, \qquad v_\rho := \frac{u_\rho}{\sqrt{a}\,L}.$$

The simple-zero part of $\widehat{A}$ is

$$P_1 = VV^{\mathsf{T}} = \sum_{\rho\in\mathcal{S}_1} v_\rho v_\rho^{\mathsf{T}},$$

where the columns of V are the v_ρ. By Lemma 2.2 of [1], $\|v_\rho\| \le 1$. For

$$Q' := \widehat{A} - P_1,$$

the decomposition in the proof of Proposition 4.4 of [1] gives

$$n_+(Q') \le s_2 + p. \tag{2}$$

Set

$$H_{\mathrm{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036794116457\ldots.$$

Theorem D of [1] proves the lower bound H_{MT}. We prove the following refinement.

Theorem 1.1. *One has*

$$\liminf_{T\to\infty} \frac{N_0^s(T,2T)}{N(T,2T)} \ge \frac{1{,}345{,}000 H_{\mathrm{MT}} - 2{,}680}{1{,}340{,}003} = 0.6730085279277797613\ldots.$$

2 A stability refinement of the rank–trace inequality

Define the convex function

$$\Psi(t) := \begin{cases} (t-1)^2, & 0 \le t \le 2, \\ 2t-3, & t \ge 2. \end{cases} \tag{3}$$

For a positive semidefinite matrix M, let

$$\mathcal{D}(M) := \operatorname{tr} \Psi(M).$$

Lemma 2.1 (Stability-enhanced rank–trace inequality)**.** *Let V be a $d \times r$ matrix whose columns have norm at most one. Put $P = VV^* \succeq 0$ and $M = V^*V$, and let Q be Hermitian with $n_+(Q) \le b$. Then*

$$\|P+Q\|_{\mathrm{F}}^2 \ge 4\operatorname{tr}(P+Q) - 3r - 4b + \mathcal{D}(M). \tag{4}$$

Proof. Write $Q = Q_+ - Q_-$ with $Q_\pm \succeq 0$, $Q_+Q_- = 0$, and $\operatorname{rank} Q_+ \le b$. Since $P, Q_+ \succeq 0$,

$$\|P+Q\|_{\mathrm{F}}^2 \ge \|P-Q_-\|_{\mathrm{F}}^2 + \|Q_+\|_{\mathrm{F}}^2.$$

If $q_j > 0$ are the nonzero eigenvalues of Q_+, then $q_j^2 \ge 4q_j - 4$, so

$$\|Q_+\|_{\mathrm{F}}^2 \ge 4\operatorname{tr} Q_+ - 4b. \tag{5}$$

Let $p_1 \ge \cdots \ge p_r \ge 0$ be the eigenvalues of M (the nonzero eigenvalues of P, padded by zeros), and let $n_1 \ge n_2 \ge \cdots \ge 0$ be the eigenvalues of Q_-. Von Neumann's trace inequality gives $\operatorname{tr}(PQ_-) \le \sum_{i\le r} p_i n_i$, hence

$$\|P-Q_-\|_{\mathrm{F}}^2 + 4\operatorname{tr} Q_- \ge \sum_{i\le r} \big((p_i - n_i)^2 + 4n_i\big).$$

For $p \ge 0$,

$$\min_{n\ge 0} \big((p-n)^2 + 4n\big) = \begin{cases} p^2, & 0 \le p \le 2, \\ 4p-4, & p \ge 2, \end{cases} = 2p - 1 + \Psi(p).$$

Therefore

$$\|P - Q_-\|_{\mathrm{F}}^2 \geq 2 \operatorname{tr} P - r + \mathcal{D}(M) - 4 \operatorname{tr} Q_-. \tag{6}$$

Combining (5) and (6) gives

$$\|P + Q\|_{\mathrm{F}}^2 \geq 4 \operatorname{tr}(P + Q) - 2 \operatorname{tr} P - r - 4b + \mathcal{D}(M).$$

Since $\operatorname{tr} P$ is the sum of the squared column norms of V, $\operatorname{tr} P \leq r$. □

Corollary 2.2. *With $M := V^{\mathsf{T}} V$,*

$$s_1 \geq 4 \operatorname{tr} \widehat{A} - \|\widehat{A}\|_{\mathrm{F}}^2 - 2N(I') + \mathcal{D}(M). \tag{7}$$

After removing the tails as in Proposition 4.2 of [1],

$$N_0^s(T, 2T) \geq H_{\mathrm{MT}} N(T, 2T) + \mathcal{D}(M^\circ) - o(N(T, 2T)), \tag{8}$$

where M° is the Gram matrix of the retained central simple zeros.

Proof. Apply Lemma 2.1 to $P_1 + Q' = \widehat{A}$. Equations (1) and (2) give

$$\|\widehat{A}\|_{\mathrm{F}}^2 \geq 4 \operatorname{tr} \widehat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \geq 4 \operatorname{tr} \widehat{A} - s_1 - 2N(I') + \mathcal{D}(M),$$

which is (7).

Use the same comparison $\widehat{A} = \widehat{G} - \widehat{E}$ and tail estimates as in Propositions 4.2 and 4.4 of [1]. Pinching the full Gram matrix into the retained central block and its complement gives $\mathcal{D}(M) \geq \mathcal{D}(M^\circ)$ because $X \mapsto \operatorname{tr} \Psi(X)$ is convex and unitarily invariant. Moreover $N(I') = N(T, 2T) + o(N(T, 2T))$, and Theorem D gives

$$\operatorname{tr} \widehat{G} = N(T, 2T)(1 + o(1)), \qquad \|\widehat{G}\|_{\mathrm{F}}^2 = \left(\frac{1}{c_1^*} + o(1)\right) N(T, 2T),$$

where $2 - 1/c_1^* = H_{\mathrm{MT}}$. Substitution in (7) yields (8). □

3 The Montgomery–Taylor overlap kernel

Put $h := 2\pi/L$ and define the normalised ordinate

$$x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

Lemma 3.1. *Fix $R_0 > 0$. Delete the simple zeros in strips of normalised width L^2 at the two ends of the grid. The number deleted is $o(N(T, 2T))$, and uniformly for retained simple zeros with $|x_\rho - x_{\rho'}| \leq R_0$,*

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1), \tag{9}$$

where

$$k(x) := \frac{K(x)}{K(0)}, \tag{10}$$

$$\begin{aligned} K(x) &:= \int_{-1/2}^{1/2} \cos(\sqrt{2}\, t) \cos(2\pi x t)\, dt \\ &= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \end{aligned} \tag{11}$$

and $K(0) = \sqrt{2} \sin(1/\sqrt{2})$. The singularities in (11) are removable.

Proof. For the full Gabor grid, Lemma 2.2 of [1] gives

$$\frac{1}{aL^2}\sum_{j\in\mathbb{Z}}\widehat{\phi}(\gamma-\tau_j)\widehat{\phi}(\gamma'-\tau_j)=\frac{\Phi(\gamma-\gamma')}{aL},\qquad \Phi=\widehat{\phi^2}.$$

For points at normalised distance at least L^2 from the grid ends, the terms with $j\notin[0,d)$ contribute $O(L^{-2})$, uniformly when $|x_\rho - x_{\rho'}|$ is bounded, by the r^{-2} decay used in [1, §5.3].

For fixed x, substitution $u = Lt$ gives the normalised full-grid overlap

$$\frac{\Phi(hx)}{aL}=\frac{\displaystyle\int_{-1/2}^{1/2}\phi(Lt)^2e^{-2\pi ixt}\,dt}{\displaystyle\int_{-1/2}^{1/2}\phi(Lt)^2\,dt}\longrightarrow\frac{K(x)}{K(0)}. \tag{12}$$

Indeed, $\phi(Lt)^2\to\cos(\sqrt{2}\,t)\mathbf{1}_{[-1/2,1/2]}(t)$ in L^1; the fixed-width taper has rescaled length $O(L^{-1})$. The convergence in (12) is uniform for x in compact sets. Together with the tail estimate, this proves (9). Finally, the deleted strips have ordinate length $O(L)$ and contain $O(L^2)=o(N(T,2T))$ zeros, since $N(t+1)-N(t)\ll\log(t+3)$. □

Set

$$w(x) := k(x)^2.$$

4 A seven-point local inequality

For nonnegative gaps $g_1,\dots,g_6$, define

$$\mathcal{F}_6(g_1,\dots,g_6):=\frac{1}{3000}\sum_{i=1}^{6}g_i+\sum_{r=1}^{6}\frac{2}{7-r}\sum_{i=1}^{7-r}w(g_i+\cdots+g_{i+r-1}). \tag{13}$$

The second term contains all 21 pairwise separations among seven ordered points.

Proposition 4.1 (Certified local inequality)**.** *For every $g_1,\dots,g_6\ge 0$,*

$$\mathcal{F}_6(g_1,\dots,g_6)\ge\frac{19}{5000}. \tag{14}$$

Verification. The verifier at

`https://github.com/ainta/zeta-simple-zeros`

uses Arb interval arithmetic through `python-flint`. It evaluates the entire-sinc expression

$$K(x)=\frac{1}{2}\left(\frac{\sin(\pi x-1/\sqrt{2})}{\pi x-1/\sqrt{2}}+\frac{\sin(\pi x+1/\sqrt{2})}{\pi x+1/\sqrt{2}}\right)$$

at 128-bit precision and constructs rigorous lower bounds for w on a grid of mesh 1/4000. If $\sum g_i\ge 11.4$, the linear term proves the claim. On the remaining compact region, the one-variable term

$$U(g):=\frac{g}{3000}+\frac{1}{3}w(g)$$

reduces each coordinate to three certified unions of grid cells. The resulting 3^6 boxes are proved by interval lower bounds, a certified convex tangent bound when applicable, or bisection of the widest coordinate. The program fails if a terminal box remains unresolved.

The committed certificate records `verified=true`, grid 4000, precision 128 bits, 707,901 visited nodes, and maximum depth 37. It also records the hashes of the interval tables used in the search. ☐

Lemma 4.2. *Let $y_1 < \cdots < y_m$ with $m \geq 7$, and put*

$$E_m := 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i).$$

Then

$$E_m + \frac{1}{500}(y_m - y_1) \geq \frac{19}{5000}(m - 6). \tag{15}$$

Proof. Write $g_i = y_{i+1} - y_i$ and sum (14) over the $m - 6$ consecutive seven-point windows. A pair spanning r gaps occurs at most $7 - r$ times and has coefficient $2/(7 - r)$ each time, so its total contribution is at most $2w(y_j - y_i)$. Each single gap occurs at most six times, giving at most

$$\frac{6}{3000} \sum_{i=1}^{m-1} g_i = \frac{1}{500}(y_m - y_1).$$

☐

Lemma 4.3. *If $G \succeq 0$ is Hermitian, then*

$$\operatorname{tr} \Psi(G) \geq \min \left\{ 1,\ 2 \sum_{i<j} |G_{ij}|^2 \right\}. \tag{16}$$

Proof. If every eigenvalue of G is at most 2, then $\Psi(G) = (G - I)^2$, and

$$\operatorname{tr} \Psi(G) = \operatorname{tr}(G - I)^2 \geq 2 \sum_{i<j} |G_{ij}|^2.$$

If some eigenvalue $\lambda > 2$, then $\Psi(\lambda) = 2\lambda - 3 > 1$. ☐

Take

$$m := 269, \qquad A_0 := \frac{19}{5000}(m - 6) = \frac{4997}{5000} < 1. \tag{17}$$

Let B be a block of 269 consecutive retained simple zeros. Write G_B for its Gram matrix. If $\operatorname{span}(B)/500 \geq A_0$, then (18) below is immediate. Otherwise $\operatorname{span}(B) < 500$, so Lemma 3.1 applies uniformly to every pair in B. Since m is fixed,

$$2 \sum_{i<j} |(G_B)_{ij}|^2 = E_m + o(1).$$

Lemmas 4.2 and 4.3 give, uniformly in B,

$$\mathcal{D}(G_B) + \frac{1}{500} \operatorname{span}(B) \geq A_0 - o(1). \tag{18}$$

5 Shifted block pinching

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalised ordinates of the retained central simple zeros. Then

$$S^\circ = N_0^s(T, 2T) - o(N(T, 2T)). \tag{19}$$

For each of the $m = 269$ offsets, partition the ordered points into full consecutive blocks of size m, leaving at most $2(m-1)$ endpoint points. Pinching M° to the full principal blocks and the remaining block gives

$$\mathcal{D}(M^\circ) \geq \sum_B \mathcal{D}(G_B). \tag{20}$$

This follows because pinching is an average of unitary conjugations and $X \mapsto \operatorname{tr}\Psi(X)$ is convex and unitarily invariant.

For each offset, sum (18) over the full blocks and use (20); then average over the m offsets. The average number of full blocks is $S^\circ/m + O(1)$. Each gap $x_{j+1} - x_j$ belongs to a block span for at most $m-1$ offsets, and

$$x_{S^\circ} - x_1 \leq \frac{T}{h} = \frac{LT}{2\pi} = d + O(1) = N(T, 2T) + o(N(T, 2T))$$

by the Riemann–von Mangoldt formula. The blockwise $o(1)$ term in (18) is uniform, so its total is $o(N(T, 2T))$. Therefore

$$\mathcal{D}(M^\circ) \geq \frac{A_0}{m} N_0^s(T, 2T) - \frac{m-1}{500m} N(T, 2T) - o(N(T, 2T)). \tag{21}$$

With $m = 269$ and $A_0 = 4997/5000$,

$$\mathcal{D}(M^\circ) \geq \frac{4997}{1{,}345{,}000} N_0^s(T, 2T) - \frac{268}{134{,}500} N(T, 2T) - o(N(T, 2T)). \tag{22}$$

Substitute (22) into (8). This gives

$$\begin{aligned} N_0^s(T, 2T) \geq H_{\mathrm{MT}} N(T, 2T) &+ \frac{4997}{1{,}345{,}000} N_0^s(T, 2T) \\ &- \frac{268}{134{,}500} N(T, 2T) - o(N(T, 2T)). \end{aligned}$$

Hence

$$\left(1 - \frac{4997}{1{,}345{,}000}\right) N_0^s(T, 2T) \geq \left(H_{\mathrm{MT}} - \frac{268}{134{,}500}\right) N(T, 2T) - o(N(T, 2T)).$$

Divide by $N(T, 2T)$ and let $T \to \infty$ to obtain

$$\liminf_{T \to \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{1{,}345{,}000 H_{\mathrm{MT}} - 2{,}680}{1{,}340{,}003} = 0.6730085279277797613\ldots.$$

This proves Theorem 1.1.

Reproducibility

Proposition 4.1 is the only computer-assisted input. Its source, tests, trust-base description, and certificate are available at

`https://github.com/ainta/zeta-simple-zeros`.

The verifier reconstructs every transcendental interval enclosure from the stated formulas.

References

[1] Claude, *More than two thirds of the zeros of the Riemann zeta function lie on the critical line*, preprint, August 10, 2026, available online.

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